

CS 8001 OIC Intro To C Programming

Spencer Collins

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## Assignment 1 Questions

### **1.1 What do you hope to learn from this course? What skills do you seek to acquire?**

I hope to learn enough about the C language to pass the Graduate Intro to Operating Systems course in Fall 2026 or Spring 2027. I plan to make my career in C/C++ low level and systems programming. I plan to be able to be comfortable with the C compiler and the ability to know the C language in technical way.

### **1.2 The spec for PSI, the language for which you'll be building an interpreter, is available in Canvas under the Files tab. Skim over it—don't spend more than an hour. You're not expected to know how to implement all these features yet, but tell me which ones you expect to be most difficult, and why.**

I am unsure which of the features would be the most difficult to implement since I have never made an interpreter before or implemented any of the features listed in the specification from scratch before.

### **1.3 What does the F in DITFOW stand for?**

The "F" in DITFOW means Familiar.

### **1.4 Type out the following. "I, <name>, have reviewed and understand the AI Policy, as expressed in Module 1."**

I, Spencer, have reviewed and understand the AI Policy, as expressed in Module 1.

## Assignment 1 Project

### **1.5 What does mystery do? Try running it for different values, starting small, to get a sense of what it's doing.**

The mystery function in hueplot.c generates a plot of the Mandelbrot set.

### **1.6 Write a program that does something—anything. Compile it, run it, and tell me how you verified that it works. Include your code with your submission.**

I have written a C program that adds, multiplies, and subtracts 45 and 56. It also will divide 9 by 3 and 9 by 0. I know it works because it returns all the correct values and compiles correctly.

Code:

```
#include <inttypes.h>
```

```
#include <math.h>
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
double add(double a, double b) {  
    return a + b;  
}
```

```
double subtract(double a, double b) {  
    return a - b;  
}
```

```
double multiply(double a, double b) {  
    return a * b;  
}
```

```
double divide(double a, double b) {  
    if (b == 0) {  
        printf("%s", "Cannot divide by ");  
        return 0.0;  
    }  
    return a / b;  
}
```

```
int main() {  
  
    printf("%f", add(45, 56));  
    printf("%s", "\n");  
    printf("%f", subtract(45, 56));  
    printf("%s", "\n");  
    printf("%f", multiply(45, 56));  
    printf("%s", "\n");  
    printf("%f", divide(9, 3));  
    printf("%s", "\n");  
    printf("%f", divide(9, 0));  
    return 0;  
};
```